Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which statements about the `final` modifier are correct? (Choose all that apply.)

A. Instance and `static` variables can be marked `final`.

B. A variable is effectively final only if it is marked `final`.

C. An object that is marked `final` cannot be modified.

D. Local variables cannot be declared with type `var` and the `final` modifier.

E. A primitive that is marked `final` cannot be modified.

2. Which of the following can fill in the blank in this code to make it compile? (Choose all that apply.)

```
public class Ant {
   _____ void method() {}
}
```

A. `default`

B. `final` Indica que el método no puede ser sobrescrito por ninguna subclase

C. `private` Restringe la visibilidad del método para que solo pueda ser llamado desde dentro de la misma clase

D. `Public`

E. `String`

F. `zzz:`

3. Which of the following methods compile? (Choose all that apply.)

A. `final static void rain() {}`

B. `public final int void snow() {}`

C. `private void int hail() {}`

D. `static final void sleet() {}`

E. `void final ice() {}`

F. `void public slush() {}`

4. Which of the following can fill in the blank and allow the code to compile? (Choose all that apply.)

```
final ________ song = 6;
```

A. `int` Tipo de dato base

B. `Integer` El autoboxing permite que su clase envoltoria castee esto

C. `long` Un int cabe dentro de un long

D. `Long`

E. `double` Un int cabe en un double, o simplemente se almacena como 6.0

F. `Double`

5. Which of the following methods compile? (Choose all that apply.)

   A. `public void january() { return; }` Sirve para detener la ejecución y salir del méotod en cualquier punto

   B. `public int february() { return null;}`

   C. `public void march() {}` Es válido, ya que no se espera un valor de retorno.

   D. `public int april() { return 9;}` Es el caso ideal para retornar un valor

   E. `public int may() { return 9.0;}`

   F. `public int june() { return;}`

6. Which of the following methods compile? (Choose all that apply.)

   A. `public void violin(int... nums) {}` Forma tradicional de agregar varargs

   B. `public void viola(String values, int... nums) {}` Los varargs deben ir en segundo lugar del argumento

   C. `public void cello(int... nums, String values) {}`

   D. `public void bass(String... values, int... nums) {}`

   E. `public void flute(String[] values, ...int nums) {}`

   F. `public void oboe(String[] values, int[] nums) {}` Es correcto, las varargs sí pueden tener arrays e ir en segunda posición.

7. Given the following method, which of the method calls return 2? (Choose all that apply.)

```
public int juggle(boolean b, boolean... b2) {
   return b2.length;
}
```

   A. `juggle();`

   B. `juggle(true);`

   C. `juggle(true, true);`

   D. `juggle(true, true, true);` Es la forma "tradicional" de pasar valores en el argumento.

   E. `juggle(true, {true, true});`

   F. `juggle(true, new boolean[2]);` Es válido colocar un array, ya que sí agregas una o más variables

8. Which of the following statements is correct?

   A. Package access is more lenient than `protected` access.

   B. A `public` class that has private fields and package methods is not visible to classes outside the package.

   C. You can use access modifiers so only some of the classes in a package see a particular package class.

   D. You can use access modifiers to allow access to all methods and not any instance variables.

   E. You can use access modifiers to restrict access to all classes that begin with the word `Test`.

9. Given the following class definitions, which lines in the `main()` method generate a compiler error? (Choose all that apply.)

```
// Classroom.java
package my.school;
public class Classroom {
   private int roomNumber;
   protected static String teacherName;
   static int globalKey = 54321;
   public static int floor = 3;
   Classroom(int r, String t) {
      roomNumber = r;
      teacherName = t; } }

// School.java
1: package my.city;
2: import my.school.*;
3: public class School {
4:    public static void main(String[] args) {
5:       System.out.println(Classroom.globalKey);
6:       Classroom room = new Classroom(101, "Mrs. Anderson");
7:       System.out.println(room.roomNumber);
8:       System.out.println(Classroom.floor);
9:       System.out.println(Classroom.teacherName); } }
```

A. None: the code compiles fine.

B. Line 5

C. Line 6

D. Line 7

E. Line 8 floor es publico, quiere decir que puede accederse desde cualquier paquete del proyecto

F. Line 9

10. What is the output of executing the `Chimp` program?

```
// Rope.java
1: package rope;
2: public class Rope {
3:    public static int LENGTH = 5;
4:    static {
5:       LENGTH = 10;
6:    }
7:    public static void swing() {
```

```
8:        System.out.print("swing ");
9:     } }
```

```
// Chimp.java
1: import rope.*;
2: import static rope.Rope.*;
3: public class Chimp {
4:    public static void main(String[] args) {
5:       Rope.swing();
6:       new Rope().swing();
7:       System.out.println(LENGTH);
8:    } }
```

A. `swing swing 5`

B. `swing swing 10` Se inicializa en 5, pero al ejecutar el bloque de Rope lo cambia a 10.

C. Compiler error on line 2 of `Chimp`

D. Compiler error on line 5 of `Chimp`

E. Compiler error on line 6 of `Chimp`

F. Compiler error on line 7 of `Chimp`

11. Which statements are true of the following code? (Choose all that apply.)

```
1:  public class Rope {
2:     public static void swing() {
3:        System.out.print("swing");
4:     }
5:     public void climb() {
6:        System.out.println("climb");
7:     }
8:     public static void play() {
9:        swing();
10:       climb();
11:    }
12:    public static void main(String[] args) {
13:       Rope rope = new Rope();
14:       rope.play();
15:       Rope rope2 = null;
16:       System.out.print("-");
17:       rope2.play();
18:    } }
```

A. The code compiles as is.

B. There is exactly one compiler error in the code.

C. There are exactly two compiler errors in the code.

D. If the line(s) with compiler errors are removed, the output is `swing-climb`.

E. If the line(s) with compiler errors are removed, the output is `swing-swing`. rope2 es null, play es estático, y solo se imprime swing

F. If the line(s) with compile errors are removed, the code throws a `NullPointerException`.

12. How many variables in the following method are effectively final?

```
10: public void feed() {
11:    int monkey = 0;
12:    if(monkey > 0) {
13:       var giraffe = monkey++;
14:       String name;
15:       name = "geoffrey";
16:    }
17:    String name = "milly";
18:    var food = 10;
19:    while(monkey <= 10) {
20:       food = 0;
21:    }
22:    name = null;
23: }
```

A. 1

B. 2 giraffe y name

C. 3

D. 4

E. 5

F. None of the above. The code does not compile.

13. What is the output of the following code?

```
// RopeSwing.java
import rope.*;
import static rope.Rope.*;
public class RopeSwing {
   private static Rope rope1 = new Rope();
   private static Rope rope2 = new Rope();
   {
      System.out.println(rope1.length);
   }
```

```
   public static void main(String[] args) {
      rope1.length = 2;
      rope2.length = 8;
      System.out.println(rope1.length);
   }
}

// Rope.java
package rope;
public class Rope {
   public static int length = 0;
}
```

A. 02

B. 08

C. 2

D. 8 rope2.lenght = 8 además es static.

E. The code does not compile.

F. An exception is thrown.

14. How many lines in the following code have compiler errors?

```
1:  public class RopeSwing {
2:     private static final String leftRope;
3:     private static final String rightRope;
4:     private static final String bench;
5:     private static final String name = "name";
6:     static {
7:        leftRope = "left";
8:        rightRope = "right";
9:     }
10:    static {
11:       name = "name";
12:       rightRope = "right";
13:    }
14:    public static void main(String[] args) {
15:       bench = "bench";No se puede inicializar en un método main
16:    }
17: }
```

A. 0

B. 1

C. 2

D. 3

E. 4

F. 5

15. Which of the following can replace line 2 to make this code compile? (Choose all that apply.)

```
1: import java.util.*;
2: // INSERT CODE HERE
3: public class Imports {
4:    public void method(ArrayList<String> list) {
5:       sort(list);
6:    }
7: }
```

A. `import static java.util.Collections;`

B. `import static java.util.Collections.*;` Se llama a la biblioteca de donde proviene esa función

C. `import static java.util.Collections.sort(ArrayList<String>);`

D. `static import java.util.Collections;`

E. `static import java.util.Collections.*;`

F. `static import java.util.Collections.sort(ArrayList<String>);`

16. What is the result of the following statements?

```
1:  public class Test {
2:     public void print(byte x) {
3:        System.out.print("byte-");
4:     }
5:     public void print(int x) {
6:        System.out.print("int-");
7:     }
8:     public void print(float x) {
9:        System.out.print("float-");
10:    }
11:    public void print(Object x) {
12:       System.out.print("Object-");
13:    }
14:    public static void main(String[] args) {
15:       Test t = new Test();
16:       short s = 123;
17:       t.print(s);      Busca la coincidencia más cercana que es un int, print(int x)
18:       t.print(true);   La coincidencia más cercana es Object x
```

```
19:         t.print(6.789);
20:     }
21: }
```

Double no se puede encojer a float, así que su coincidencia más cercana hacia arriba es object

A. `byte-float-Object-`

B. `int-float-Object-`

C. `byte-Object-float-`

D. `int-Object-float-`

E. `int-Object-Object-`

F. `byte-Object-Object-`

17. What is the result of the following program?

```
1:  public class Squares {
2:     public static long square(int x) {
3:        var y = x * (long) x;
4:        x = -1;
5:        return y;
6:     }
7:     public static void main(String[] args) {
8:        var value = 9;
9:        var result = square(value);
10:       System.out.println(value);
11:    } }
```

A. `-1`

B. `9` El código imprime value y no result

C. `81`

D. Compiler error on line 9

E. Compiler error on a different line

18. Which of the following are output by the following code? (Choose all that apply.)

```
public class StringBuilders {
   public static StringBuilder work(StringBuilder a,
      StringBuilder b) {
      a = new StringBuilder("a");
      b.append("b");
      return a;
   }
   public static void main(String[] args) {
      var s1 = new StringBuilder("s1");
      var s2 = new StringBuilder("s2");
```

```
        var s3 = work(s1, s2);
        System.out.println("s1 = " + s1);
        System.out.println("s2 = " + s2);
        System.out.println("s3 = " + s3);
    }
}
```

A. `s1 = a`

B. `s1 = s1`

C. `s2 = s2`

D. `s2 = s2b`

E. `s3 = a`

F. The code does not compile.

19. Which of the following will compile when independently inserted in the following code? (Choose all that apply.)

```
1:  public class Order3 {
2:     final String value1 = "red";
3:     static String value2 = "blue";
4:     String value3 = "yellow";
5:     {
6:        // CODE SNIPPET 1
7:     }
8:     static {
9:        // CODE SNIPPET 2
10:    } }
```

A. Insert at line 6: `value1 = "green";`

B. Insert at line 6: `value2 = "purple";`

C. Insert at line 6: `value3 = "orange";`

El bloque de instancia puede acceder y modificar variables estáticas

D. Insert at line 9: `value1 = "magenta";`

E. Insert at line 9: `value2 = "cyan";`

F. Insert at line 9: `value3 = "turquoise";`

20. Which of the following are true about the following code? (Choose all that apply.)

```
public class Run {
   static void execute() {
      System.out.print("1-");
   }
   static void execute(int num) {
      System.out.print("2-");
   }
```

```
    static void execute(Integer num) {
        System.out.print("3-");
    }
    static void execute(Object num) {
        System.out.print("4-");
    }
    static void execute(int... nums) {
        System.out.print("5-");
    }
    public static void main(String[] args) {
        Run.execute(100);
        Run.execute(100L);
    }
}
```

A. The code prints out 2-4-.

Agregar 100 incurre en un int, e imprime 2-.
Posteriormente al no haber un long, se recurre a los que sea más grande, como no hay finalmente cae en un object.

B. The code prints out 3-4-.

C. The code prints out 4-2-.

D. The code prints out 4-4-.

E. The code prints 3-4- if you remove the method `static void execute(int num)`.

F. The code prints 4-4- if you remove the method `static void execute(int num)`.

21. Which method signatures are valid overloads of the following method signature? (Choose all that apply.)

```
public void moo(int m, int... n)
```

A. `public void moo(int a, int... b)`

B. `public int moo(char ch)` Sobrecarga el método correctamente

C. `public void moooo(int... z)`

D. `private void moo(int... x)` El modificador de acceso no influye en la sobrecargar

E. `public void moooo(int y)`

F. `public void moo(int... c, int d)`

G. `public void moo(int... i, int j...)`